Exam 1

1) i) Is $f(x) = |x-2|$ a convex function.

→ A function $f: \mathbb{R} \to \mathbb{R}$ is convex if for all $x, y \in \mathbb{R}$ & $\lambda \in [0,1]$.

→ Here, the function $f(x) = |x-2|$ is the absolute value function shifted to the right by 2.

→ Absolute value functions are V-shaped & convex.

→ Hence the answer is 'Yes' it is convex.

ii) Does it have bounded gradients:

→ Given, we need to assume gradient at $x=2$ is 0.

→ For $x < 2$, ; For $x > 2$
$f'(x) = -1$; $f'(x) = 1$

→ At $x=2$, it is not differentiable, but we assume $f'(2) = 0$ for analysis

→ We know,

$$|f'(x) - f'(y)| \leq L|x-y|$$

Let us try $x = 1.9$, $y = 2.1$

$f'(1.9) = -1$; $f'(2.1) = 1$;

⇒ $|f'(1.9) - f'(2.1)| = |-1-1| = 2$
$|1.9 - 2.1| = 0.2$

So, we get $|f'(x) - f'(y)| \leq L|x-y|$

Substituting ⇒ $2 \leq L(0.2) \Rightarrow L \geq 10$

So, for any smaller interval approaching 2, the ratio will go to infinity as denominator tends to 0 & numerator is fixed at 2.

⇒ Hence, $f(x) = |x-2|$ doesn't have bounded gradients on $\mathbb{R}$, because the derivative changes discontinuously at $x=2$; causing gradient to blow up

iii) Gradient descent from $x_0 = 1.05$ & $\eta = 0.1$

$f(x) = |x-2|$

$f'(x) = -1$ for $x < 2$

Iteration 1:

$x_1 = x_0 - \eta \cdot f'(x_0) = 1.05 - 0.1(-1) = 1.15$

$\Rightarrow x_2 = 1.25$

$\Rightarrow x_3 = 1.35$

$\vdots$

This will contine until $x_t > 2$, when gradient flips to $+1$

So, once x crosses 2, gradient flips & this causes oscillations around 2

$x_t \in \{ \dots, 1.95, 2.05, 1.95, 2.05, \dots \}$

$\Rightarrow$ So, it won't converge around global minimum at 2 & it oscillates.

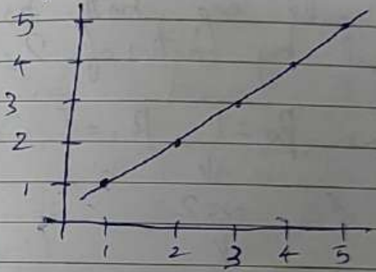
2) a) Given

$S = (x_1, y_1) \dots (x_t, y_t)$

Let us assume the data points as,

$(1,1), (2,2), (3,3), (4,4), (5,5)$

So, the regression line would be



So, here b_0 (y-intercept) will be 0 & b_1 is 1.

Now ~~let~~ us assume $\alpha = 2$; Given $y' = \alpha y$

So, the new points would be

$(1,2), (2,4), (3,6), (4,8), (5,10)$

$\Rightarrow$ This makes the slope of the line higher, but the y-intercept doesn't seem to be impacted

But if we take another example,

Initial (y) $(1,2), (2,3), (3,4), (4,5)$

$$\downarrow \alpha = 2 \downarrow$$

Final (y') $(1,4), (2,6), (3,8), (4,10)$

Here we could see both b_0 & b_1 have increased by factor of 2, as

~~$b_0 = 1$, $b_1 =$~~ $\beta_0 = 1$, $\beta_1 = 1$

$$\downarrow \alpha = 2 \downarrow$$

$\beta_0 = 2$, $\beta_1 = 2$

This concludes that, if y_is are rescaled ~~to~~ y_i's by α, then

β_0 becomes $\alpha \beta_0$ & β_1 becomes $\alpha \beta_1$ //

b) If x_is are scaled by α //

In the same way as previous example.

Initial (x_i) $(1,2), (2,3), (3,4), (4,5)$

$$\downarrow \alpha = 2 \downarrow$$

Final (x_i') $(2,2), (4,3), (6,4), (8,5)$

For Initial (x_i), $\beta_0 = 1$, $\beta_1 = 1$

$$\downarrow \alpha = 2 \downarrow$$

For Final (x_i'), $\beta_0 = 1$, $\beta_1 = \frac{1}{2}$

This concludes that, if x_is are rescaled by α, then

β_0 stays the same, but β_1 becomes $\frac{\beta_1}{\alpha}$ //

c) i) $y = \alpha x_1 + \beta x_2$

Yes, we can use linear regression to learn α & β here because $y = \alpha x_1 + \beta x_2$ is a linear 3-dimensional equation & it could be solved given xs & ys.

ii) $y = \alpha x_1^2 + \beta x_2^3$

Yes, we can use linear regression to learn α & β here as this equation is differentiable & it becomes linear & solvable for given xs & ys.

iii) $y = 2^{\alpha} x_1^{\beta}$

No, this equation is non-linear & we can't use linear regression to learn α & β values here.

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3) a) Given,

$$X = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}$$

Calculating mean of each column

Column 1 : $\mu_1 = \frac{6}{3} = 2$

Column 2 : $\mu_2 = \frac{18}{3} = 6$

So, Centered Matrix $\bar{X} = \begin{bmatrix} 1-2 & 3-6 \\ 2-2 & 6-6 \\ 3-2 & 9-6 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 0 & 0 \\ 1 & 3 \end{bmatrix}$

So, Covariance Matrix,

$\Sigma = \frac{1}{n-1} \bar{X}^T \bar{X}$ where $n=3$

$$\Sigma = \frac{1}{2} \begin{bmatrix} -1 & 0 & 1 \\ -3 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & -3 \\ 0 & 0 \\ 1 & 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$$

$\Rightarrow$ Eigen Values λ of matrix,

$\det(\Sigma - \lambda I) = 0$

$\Rightarrow \begin{bmatrix} 1-\lambda & 3 \\ 3 & 9-\lambda \end{bmatrix} \Rightarrow (1-\lambda)(9-\lambda) - 9 = 0$

$\Rightarrow \lambda^2 - 10\lambda = 0 \Rightarrow \lambda(\lambda - 10) = 0$

$\Rightarrow \lambda = 0, 10$

So, Eigenvalues are,

$\lambda_1 = 10$

$\lambda_2 = 0$

$\Rightarrow$ First Principal Component:

Find eigenvector for $\lambda = 10$. (As it is larger)

$\Rightarrow (\Sigma - 10I)v = 0$

$\begin{bmatrix} 1-10 & 3 \\ 3 & 9-10 \end{bmatrix} \Rightarrow \begin{bmatrix} -9 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$

$\Rightarrow -9v_1 + 3v_2 = 0 \Rightarrow v_2 = 3v_1$

So, eigenvector is proportional to.

$v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Normalize it.

$\|v\| = \sqrt{1^2 + 3^2} = \sqrt{10}$

So, First Principal Component $= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

The data lies exactly on a straight line $y = 3x$.

$\Rightarrow$ No. of non zero principal components $= 1$

b) Given SVD of matrix

$A = U \Sigma V^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 1 \\ 7 & 9 & 10 \end{bmatrix}$

To find, the matrix that is the best rank-1 approximation to A.

$\Rightarrow$ We know that,

$$A_1 = \sigma_1 u_1 v_1^T$$

where

$A_1 \Rightarrow$ Best rank-1 approximation to A.

$\sigma_1 \Rightarrow$ Largest singular value of A.

$u_1 \Rightarrow$ First left singular vector

$v_1 \Rightarrow$ First right singular vector

Here $\sigma_1 = 16.180$

$$u_1 = \begin{bmatrix} -0.222 \\ -0.270 \\ -0.937 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} -0.486 \\ -0.716 \\ 0.501 \end{bmatrix}$$

So,

$$A_1 = 16.18 \cdot \begin{bmatrix} -0.222 \\ -0.270 \\ -0.937 \end{bmatrix} \begin{bmatrix} -0.486 & -0.716 & 0.501 \end{bmatrix}$$

Solving this we will get best rank-1 approximation to A.

c) Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 1 \\ 7 & 9 & 10 \end{bmatrix}$

We know its SVD is,

$$A = U \Sigma V^T$$

In PCA, projecting a centered matrix A onto its first k principal components gives,

$$Z_k = A V_k$$

where V_k : matrix of first k right singular vectors.

But we need it only using U & Σ.

Since $A = U\Sigma V^T$ & V is orthonormal

$\Rightarrow AV = U\Sigma \Rightarrow Z = U\Sigma$

$\Rightarrow Z_k = U_k \Sigma_k$ //

where U_k : first k columns of U.
Σ_k : top left $k \times k$ submatrix of Σ.

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4) Given two circles (concentric) centered at origin, such that $r_1 \leq r_2$.

The point $x \in \mathbb{R}^2$ is

Positive if $r_1 \leq |x| \leq r_2$

Negative otherwise

a) PAC learning Algorithm.

~~$\Rightarrow$ Compute the norms of all positive examples in the sample.~~

Let training set be,

$S = \{(x_i, y_i)\}_{i=1}^{m}$ where $x_i \in \mathbb{R}^2, y_i \in \{0,1\}$

Let $|x|$ be the Euclidean norm (distance from origin)

$\Rightarrow$ Compute norms of all positive examples in the sample.

$d_i = \|x_i\|$ for all x_i such that $y_i = 1$

$\Rightarrow$ Define hypothesis h as two concentric circles with

$r_1 = \min d_i$; $r_2 = \max d_i$

$$\Rightarrow h(x) = \begin{cases} 1 & \text{if } r_1 \leq \|x\| \leq r_2 \\ 0 & \text{otherwise} \end{cases}$$

$\Rightarrow$ This forms smallest annulus (2 concentric circles) that contains all positive examples.

b) Bad events:

The hypothesis could fail when,

i) Missing Inner region

-> No positive samples near inner region r_1^*

-> Algo sets $r_1 > r_1^*$

-> Leads to false negatives in inner part of ring.

ii) Missing Outer region

-> Same as previous case, but near outer region r_2^*

-> Algo sets $r_2 < r_2^*$

-> Leads to false negatives in outer part of ring.

c) Sample Complexity:

We want

$$Pr[error(h) > \varepsilon] \leq \delta$$

where ε -> allowable error
δ -> allowable failure probability

We need to cover 2 regions to reduce the error. So,

the probability that none of m samples fall into a region of probability $\varepsilon/2$:

$$Pr[\text{miss edge}] \leq (1-\tfrac{\varepsilon}{2})^m \leq e^{-m\varepsilon/2}$$

Using union bound, similar to perceptron case in class.

$$Pr[\text{miss either edge}] \leq 2e^{-m\varepsilon/2}$$

So, $2e^{-m\varepsilon/2} \leq \delta \Rightarrow -m\varepsilon/2 \leq \log(\tfrac{\delta}{2})$

$$\Rightarrow m \geq \tfrac{2}{\varepsilon} \log\left(\tfrac{2}{\delta}\right)$$

This is the Sample Complexity Bound to ensure the algorithm is PAC

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5) a) Given
$h(x) = \text{sign}(x_i - t)$ (or) $h(x) = \text{sign}(t - x_i)$

for $i \in \{1, 2\}$, $t \in \{1, 2\}$
$w_1 = (1, 1, 1, \ldots 1)$

To find,

Weak learner h_1 that minimizes the weighted classification error ε_1

So, there are a total of 8 possible weak learners.

1) $h(x) = \text{Sign}(x_1 - 1) \Rightarrow \varepsilon = \frac{6}{7}$

2) $h(x) = \text{sign}(x_1 - 2) \Rightarrow \varepsilon = 3/7$

3) $h(x) = \text{sign}(x_2 - 1) \Rightarrow \varepsilon = 5/7$

4) $h(x) = \text{sign}(x_2 - 2) \Rightarrow \varepsilon = 2/7$

5) $h(x) = \text{sign}(1 - x_1) \Rightarrow \varepsilon = 1/7$

6) $h(x) = \text{sign}(2 - x_1) \Rightarrow \varepsilon = 4/7$

7) $h(x) = \text{sign}(1 - x_2) \Rightarrow \varepsilon = 2/7$

8) $h(x) = \text{sign}(2 - x_2) \Rightarrow \varepsilon = 5/7$

So, out of these, the best hypothesis for weak learner is,

$h(x) = \text{Sign}(1 - x_1)$

Error rate $E_1 = 1/7$

b) Following AdaBoost, the weight assigned to weak learner

We have $E_1 = 1/7$

$\Rightarrow \beta_1 = \frac{E_1}{1 - E_1} = \frac{1}{6} = 0.1667$

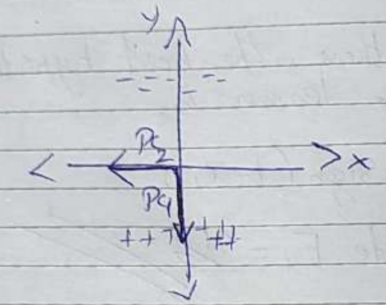
So, $w_2(i) = w_1(i) \cdot \beta_1^{1 - [h_1(x_i) = y_i]}$

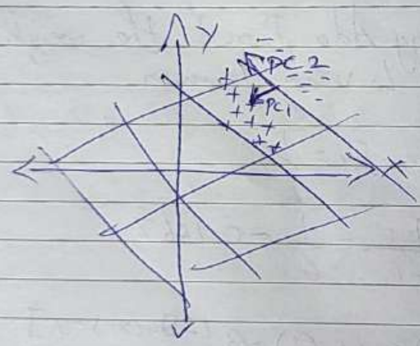
If correct, $w_2(i) = w_1(i) \cdot \beta_1^0 = w_1(i)$

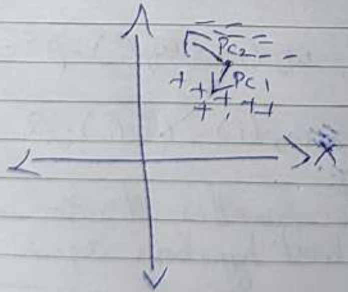
If wrong, $w_2(i) = w_1(i) \cdot \beta_1$

Based on this, another iteration could be done to find the best hypothesis again.

6) i) Principal Components







ii) Yes we can classify, if we project onto PC_1 as it completly separates the two clusters & is the ideal scenario for a PCA exercise.

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